For $x \neq -4, f(x)$ can be simplified by cancellation :

$$f(x) = \frac{16 - x^2}{4 + x}$$

$$= \frac{(4 + x)(4 - x)}{(4 + x)} = 4 - x.$$

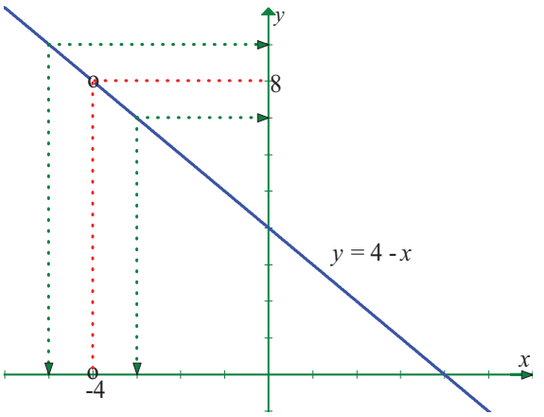


Fig. 9.2

As seen in Fig.9.2, the graph of $f(x)$ is essentially the graph of $y = 4 - x$ with the exception that the graph of f has a hole (puncture) at the point that corresponds to $x = -4$. As x gets closer and closer to -4, represented by the two arrow heads on the x-axis, the two arrow heads on the y-axis simultaneously get closer and closer to the number 8.

Here, note that

$$\lim_{x \to -4^-} f(x) = 8 = \lim_{x \to -4^+} f(x) \text{ and hence } \lim_{x \to -4} f(x) = \lim_{x \to -4} \frac{16 - x^2}{4 + x} = 8.$$

In Illustration 9.2, note that the function is not defined at $x = -4$ and yet $f(x)$ appears to be approaching a limit as x approaches -4. This often happens, and **it is important to realise that the existence or non-existence of $f(x)$ at $x = -4$ has no bearing on the existence of the limit of $f(x)$ as x approaches -4.**

Illustration 9.3

Now let us consider a function different from Illustrations 9.1 and 9.2.

Let $f(x) = \frac{|x|}{x}$.

$x = 0$ does not belong to the domain of this function, $\mathbb{R} \setminus \{0\}$. Look at the graph of this function. From the graph one can see that for positive values of x,

$$\frac{|x|}{x} = \frac{x}{x} = +1 \text{ and}$$

for negative x values, $\frac{|x|}{x} = \frac{-x}{x} = -1$.

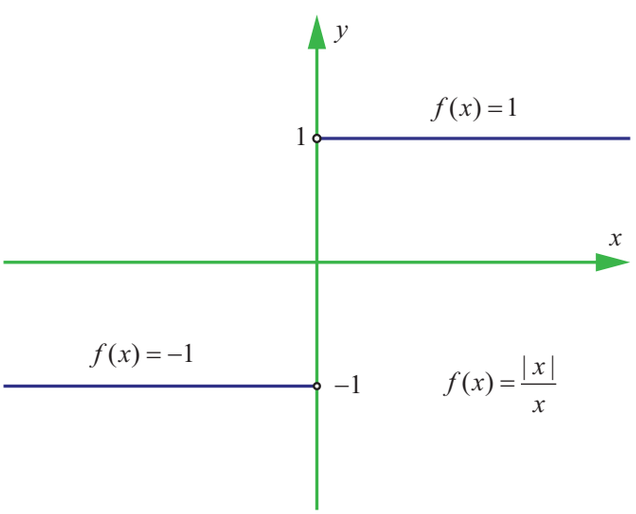


Fig. 9.3

This means that no matter how close x gets to 0 (in a small neighbourhood of 0), there will be both positive and negative x values that yield $f(x) = 1$ and $f(x) = -1$.

That is, $\lim_{x \to 0^-} f(x) = -1$ and $\lim_{x \to 0^+} f(x) = +1$.

This means that the limit does not exist. Of course, for any other value of x, there is a limit.

For example $\lim_{x \to 2^-} \frac{|x|}{x} = 1$ and $\lim_{x \to 2^+} \frac{|x|}{x} = 1$.

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